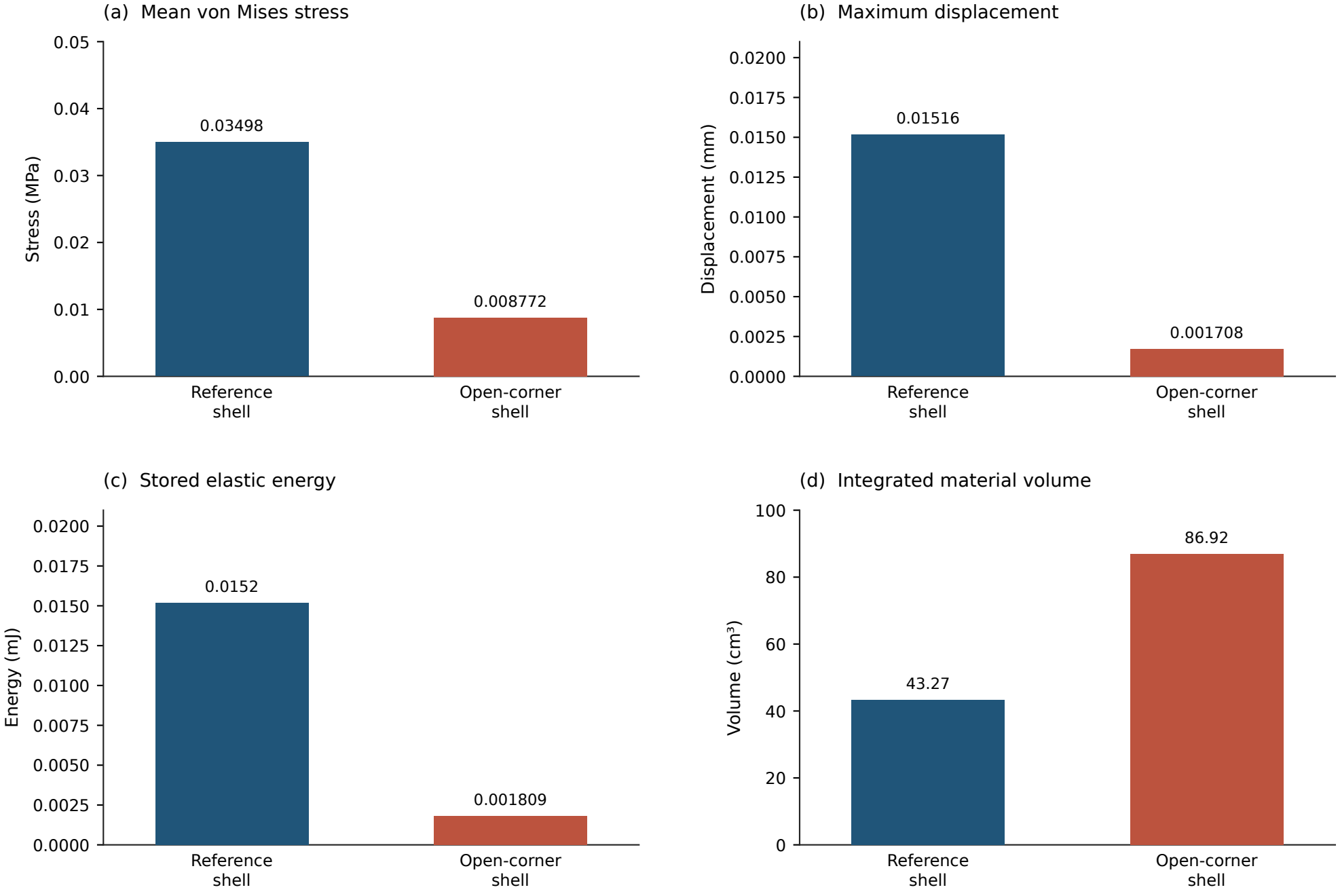
(a) Mean von Mises stress
Stress (MPa)
0.05
0.04
0.03
0.02
0.01
0.00
0.03498
0.008772
Reference shell
Open-corner shell
(b) Maximum displacement
Displacement (mm)
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
0.01516
0.001708
Reference shell
Open-corner shell
(c) Stored elastic energy
Energy (mJ)
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
0.0152
0.001809
Reference shell
Open-corner shell
(d) Integrated material volume
Volume (cm³)
100
80
60
40
20
0
43.27
86.92
Reference shell
Open-corner shell